THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May, 18, 2024 TIME: 6:30 pm - 9:30 pm

Answer ALL questions. Section A carries 30 marks and Section B carries 70 marks.

Use the same answer book for Section A and B.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (Ten One-choice questions with 3 marks each.
There will not be any deduction for wrong answers.)
<u>You may need to use Appendix A at the end of the exam paper for your calculations.</u>

1. When the volume ratio between two ideal gases (v1/v2) is 2/3, what is their molar ratio (n1/n2)?
(A) 2/3; (B) 3/2; (C) 8.31; (D) 22.4.

2. When 40,000 kJ of energy is applied to heat 0.1 m3 of cold water (25 °C) with an efficiency of 60%, how much temperature will be raised for the water?
(A) 82 °C; (B) 75 °C; (C) 60 °C; (D) 57 °C.

3. In a carbonate system, which of below is the main species when the PH = 8?
(A) H_2CO_3; (B) HCO_3^-; (C) CO_3^{2-}; (D) H_3O^+

4. Which of the following statements regarding water is **True**?
(A) Oceans, which occupy around 96.5% of earth's total water, is considered as surface water;
(B) Hydrogen bonding is a type of dipole-dipole attraction and only exist in water molecules;
(C) Freshwater occupy around 2.5% of earth's total water, and can be used by people;
(D) Temperature could be one of biological pollutants in water due to the effect on aquatic life.

5. Which of the following statements is **NOT** true for DO?
(A) DO is required for aerobic biological treatment processes in water;
(B) Higher DO at higher temperature;
(C) Low DO at higher salinity;
(D) Higher DO at high pressure.

6. In BOD_5 determinate of sewage, a 10 mL sewage mixed with enough water to fill a 400 mL bottle. The initial dissolved oxygen is measured to be 6.5 mg/L. However, after 5-days incubation at 20 °C, the dissolved oxygen is decreased 2.5 mg/L. The BOD_5 of the sewage would be
(A) 100 mg/L; (B) 160 mg/L; (C) 260 mg/L; (D) 300 mg/L.

7. Which of the following statements regarding disinfection is **True** in water treatment?
(A) Aluminum sulfate, $FeCl_3$, and $FeSO_4$ are the common disinfectant;
(B) The objective for disinfection is only to kill pathogens in water;
(C) The typical filtration rates is 5 to 25 m/hr;
(D) Disinfection at slightly acidic pH, the disinfection power of HOCl is much lower than that of OCl^-.

8. Which of the following is the highest removal efficiency during primary treatment of wastewater?
(A) Nutrient; (B) BOD; (C) TDS; (D) TSS.

9. Which of the following pollutants is not a major contributor to acid rain formation?
(A) Sulfur dioxide (SO_2); (B) Nitric oxide (NO); (C) Carbon monoxide (CO); (D) Nitric acid (HNO)

10. Which of the following is not a greenhouse gas?
(A) Argon (Ar); (B) Chlorofluorocarbons (CFCs); (C) Nitrous oxide (N_2O); (D) Water (H_2O)

Section B. (Four questions with unequal marks)

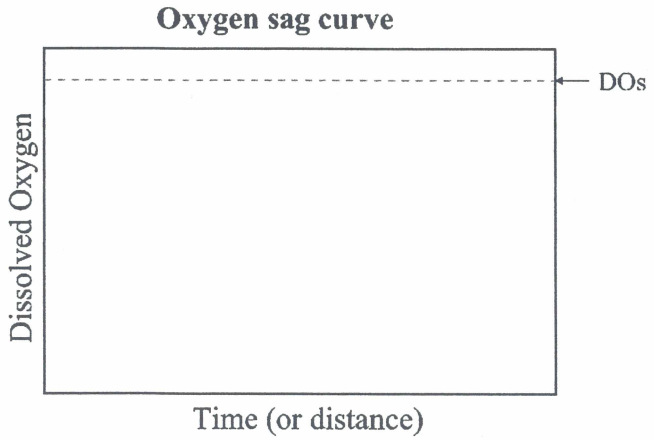
You may need to use Appendix A at the end of the exam paper for your calculations.

11. Two streams flow into a complete-mix reactor. One contains 5 ppm pollutants with a flow rate of 10 m^3/s and the other contains 50 ppm pollutants with a flow rate of 2 m^3/s. The decay rate coefficient of the pollutants is 0.5/day. Assuming the steady-state pollutant concentration in the reactor.

(1) Perform the mass balance for pollutants and water. (5 marks)

(2) How to design the volume of this reactor if the effluent water quality is required to contain pollutants less than 1 ppm? (15 marks)

12. (1) The wastewater has a certain DO concentration and discharge into the river. The river with a certain flow rate that is receiving this wastewater (Keep the temperature at 20 °C). Assuming complete and instantaneous mixing, please complete the following picture:
(Please draw the picture and your answer in the exam answer book)



i) Please draw out the oxygen sag curve, and remaining OD curves of the deoxygenation and reaeration;
ii) Please point out the minimum DO and oxygen deficit in oxygen sag curve;
iii) Please explain the phenomena of deoxygenation and reaeration;
iv) Please explain the rate of deoxygenation, the rate of reaeration, and the rate of DO increase of deficit. *Note: DOs is the saturated value of DO.*

(10 marks)

(2) Biological treatment kinetics: A well-mixed aeration lagoon rectangular has 900 m^3 volume, and it receives 300 m^3 wastewater for treatment every day. To test the BOD_5 of wastewater, the dilution factor for an unseeded mixture of wastewater and pure water is 0.035. The DO of the mixture is initially 10.0 mg/L, and 5 days later, it drops to 3.0 mg/L. What's the efficiency of this lagoon? (The biodegradation rate constants for the wastewater include endogenous decay constant k_d = 0.08 d^{-1}, maximum specific microbial growth rate μ_m = 1.10 d^{-1}, and the half velocity constant K_s = 76.0 mg BOD_5/L).

(9 marks)

13. (1) Lithium-ion battery recycling is crucial for both environmental and economic reasons. The spent lithium-ion batteries can be dissolved into an acidic solution and adjust the pH to

obtain its leaching solution. However, there is lack of efficient Li^+ separation from the high concentration competing metal ions (including, Ni^{2+}, Co^{2+} and Mn^{2+}). Membrane technology with well-defined pore structure and surface charge properties has shown great potential in achieving efficient ion separation. Briefly design an appropriate treatment process (with membrane technology) for Li^+ sustainable recovery from the spent lithium-ion batteries and explain your design purposes (**< 200 words including explanation of your design purposes, up to 1 Figure is allowed**).

(11 marks)

(2) Discharged hospital wastewater contains various pathogenic microorganisms, antibiotic groups, toxic organic compounds, radioactive elements, and ionic pollutants. These contaminants harm the environment and human health causing the spread of disease. Thus, effective treatment of hospital wastewater is an urgent task for sustainable development. Please design an appropriate treatment process, and explain your design purposes. (**< 150 words including explanation of your design purposes, up to 1 Figure is allowed**).

(10 marks)

14. Consider the phenomenon of ocean acidification, where increased levels of CO2 in the atmosphere lead to greater CO_2 absorption by the ocean, thus lowering its pH.

(1) Describe the chemical process of how carbon dioxide gas affects ocean chemistry.

(2) Discuss the ecological impacts of ocean acidification on marine life, particularly focusing on coral reefs.

(3) Propose potential strategies that could be implemented to mitigate the effects of ocean acidification on these marine ecosystems.

(4) Besides CO_2, other air pollutants like PM2.5 pose direct threats to human health. Explain why PM2.5 is considered more hazardous than PM10.

(10 marks)

Appendix A. Information for Reference

I. Atomic weights

H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

II. Properties of water

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

III. Other constants

Ideal gas law constant = 0.082056 L·atm/(K·mol)
Partial pressure of CO_2 in atmosphere: $1 \times 10^{-3.5}$ atm
Henry's Law constant for CO_2: 0.03162 mol/(L·atm)
Acidity constants for H_2CO_3: $pK_1 = 6.3$; $pK_2 = 10.3$
Solubility of $CaCO_3$(s): $pK_{so} = 8.3$
Henry's Law constant for O_2 (25 °C) = 0.001263 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1 \times 10^{-14}$
Equivalent Weight (EW) for Ca^{2+} = 20 mg/meq
EW for Mg^{2+} = 12.2 mg/meq
EW for $CaCO_3$ = 50 mg/meq

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